Cross-Correlation and FFT Phase Analysis for Defect Detection in Active Thermography of FRP Composites

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***Abstract*—We present a per-pixel signal processing pipeline for detecting subsurface defects in fibre-reinforced polymer (FRP) composites using active thermography. For each pixel in a recorded thermal video, the temporal intensity signal is linearly detrended and then cross-correlated with a reference pixel located in a sound (defect-free) region. The central main lobe of the resulting cross-correlation is extracted, and the phase of its discrete Fourier transform (DFT) is computed. Experiments are conducted on Carbon Fibre-Reinforced Polymer (CFRP) specimens under Linear Frequency Modulated (LFM) excitation and Glass Fibre-Reinforced Polymer (GFRP) specimens under active pulsed and flux excitation. For flux-excited GFRP specimens the FFT phase of the extracted correlation window produces clearly distinct spatial signatures that localise subsurface defects, while active pulsed excitation is shown to be insufficient for phase-based discrimination with this pipeline. The results demonstrate the potential of the approach — and its sensitivity to excitation waveform — for non-destructive evaluation of composite structures.**

***Index Terms*—active thermography, non-destructive testing, fibre-reinforced polymer, cross-correlation, FFT phase, defect detection, LFM excitation**

I. Introduction

Fibre-reinforced polymer (FRP) composites are widely used in aerospace, civil, and mechanical engineering due to their high strength-to-weight ratio. Subsurface manufacturing defects such as delaminations and voids can significantly compromise structural integrity, making non-destructive testing (NDT) an essential part of quality assurance.

Active thermography is an established NDT technique in which an external heat source excites the specimen and the resulting surface thermal response is recorded by an infrared camera [1]. Subsurface defects alter local heat diffusion, producing surface thermal anomalies that are detectable in the recorded video. Among the available excitation strategies, Linear Frequency Modulated (LFM) thermal excitation has attracted attention because the chirp waveform allows depth resolution to be traded against acquisition time [2]. Infrared thermal wave imaging has been applied to FRP composites under a range of such excitation strategies, demonstrating effective defect detection in both CFRP and GFRP specimens [3], [4].

A variety of signal processing methods have been applied to thermographic sequences, including pulsed phase thermography, thermographic signal reconstruction, and principal component analysis. Cross-correlation has also been explored as a means of thermally characterising CFRP materials through a pulse-compression framework [5]. In this work, we investigate a pipeline that combines per-pixel linear detrending, full cross-correlation with a reference signal, main lobe extraction, and DFT phase computation. The method is applied to three datasets spanning two material systems and two excitation modalities.

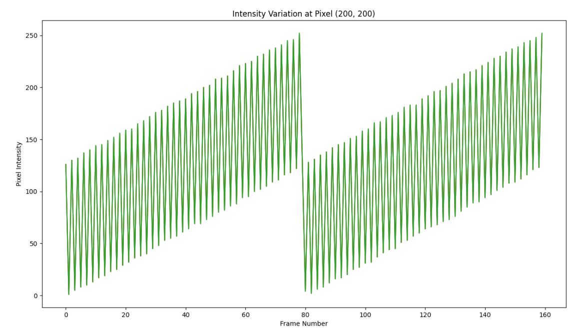

Fig. 1. Temporal intensity signal at pixel $(200, 200)$ in the GFRP flux dataset (first 180 of 1010 frames shown). Two repeating cycles of increasing-frequency oscillation are visible, consistent with the flux excitation waveform.

II. Experimental Datasets

Three thermographic video datasets are used in this study.

- **CFRP LFM.** A Carbon Fibre-Reinforced Polymer specimen excited with a Linear Frequency Modulated thermal stimulus. The video has a spatial resolution of 169×289 pixels, a frame rate of 25 fps, and a total duration of 40 s ($\approx$1000 frames).
- **GFRP AP.** A Glass Fibre-Reinforced Polymer specimen under Active Pulsed excitation, recorded at a spatial resolution of approximately 460×460 pixels over 1010 frames at 25 fps.
- **GFRP Flux.** A GFRP specimen under flux excitation, recorded at a spatial resolution of approximately 572×458 pixels over 1010 frames at 25 fps.

All videos are acquired in greyscale. Fig. 1 shows the temporal intensity profile of pixel $(200, 200)$ from the GFRP flux dataset over the first 180 of 1010 frames, illustrating two repeating cycles of an amplitude-modulated oscillation with increasing intra-cycle frequency, consistent with the flux excitation waveform.

III. Signal Processing Methodology

The processing pipeline is applied independently and identically to every spatial pixel. Fig. 2 gives an example of the intermediate cross-correlation output.

A. Greyscale Normalisation

Each video frame is converted to greyscale and cast to a 64-bit floating-point representation normalised to the range $[0, 1]$.

B. Linear Detrending

Let $I(x, y, t)$ denote the normalised intensity of pixel (x, y) at frame t. A linear trend is estimated for each pixel via least-squares regression:

$$\hat{I}(x, y, t) \;=\; \alpha\, t + \beta, \tag{1}$$

with slope α and intercept β fit to the full temporal sequence. The detrended signal is

$$\tilde{I}(x, y, t) \;=\; I(x, y, t) - \hat{I}(x, y, t). \tag{2}$$

This step removes slow bulk-heating drift, leaving the thermally-driven response to the excitation waveform.

C. Cross-Correlation with a Reference Pixel

Each detrended signal $\tilde{I}(x, y, \cdot)$ is cross-correlated with the signal at a designated reference pixel (x_0, y_0) located in a sound region:

$$R_{xy}(\tau) \;=\; \sum_{t=0}^{N-1} \tilde{I}(x, y, t)\, \tilde{I}(x_0, y_0, t-\tau), \tag{3}$$

where samples outside $[0, N-1]$ are treated as zero (zero-extension). The full cross-correlation is computed using `numpy.correlate` in `full` mode (GFRP AP, GFRP flux) or `same` mode (CFRP LFM), yielding an output of length $2N-1$ or N respectively. The `same` mode was used for the CFRP LFM dataset to match the output length to the input signal length; `full` mode was used for the GFRP datasets to retain all lag values. No coefficient normalisation is applied; raw cross-correlation values are retained. For the CFRP LFM, GFRP AP, and GFRP flux datasets the reference pixel is located at coordinates $(100, 100)$, $(200, 200)$, and $(200, 200)$ respectively, all chosen from visually defect-free regions of the specimen. This approach is related to the pulse-compression framework applied to CFRP characterisation in [5], but operates directly on a per-pixel reference signal rather than a synthetic matched filter.

D. Correlation Window Extraction

A fixed temporal window of the cross-correlation is extracted for further analysis. Both GFRP datasets comprise 1010 frames, giving a full-mode correlation output of $2\times1010-1=2019$ frames with the zero-lag peak at index 1009. The GFRP flux window (frames 930–1090, 160 frames) is centred on this zero-lag peak and constitutes the main lobe extraction. For the CFRP LFM dataset ($N=1001$, `same` mode, output length 1001) the zero-lag peak falls at index 500; a 10-frame window (frames 430–440, centred at index 435, approximately 65 samples before zero-lag) was selected by inspection of single-pixel correlation traces. For the GFRP AP dataset the

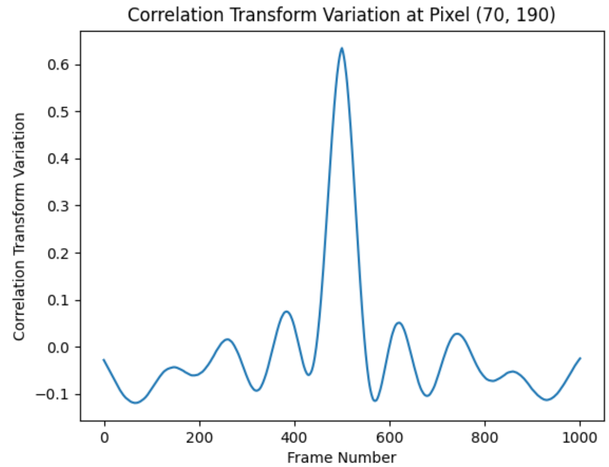

Fig. 2. Cross-correlation of pixel $(70, 190)$ with the reference pixel in the CFRP LFM dataset. A distinct main lobe is centred at zero lag, with decaying side lobes on either side.

window spans frames 400–610 (210 frames of the 2019-frame output, centred at index 505), which lies approximately 500 samples before the zero-lag peak at index 1009; this off-centre placement reflects the empirical response structure of the active-pulsed excitation and is consistent with the negative phase-separation result discussed in Section IV.

E. FFT Phase Computation

The discrete Fourier transform is applied to the extracted correlation window, and its phase spectrum is retained:

$$\phi_{xy}(k) \;=\; \angle\, \mathcal{F}\{R_{xy}(\tau)\}\,[k], \tag{4}$$

where k indexes frequency bin. This per-pixel phase sequence constitutes the primary output used for defect discrimination.

F. Discretisation (Zero-Padding)

To investigate the effect of frequency resolution on the phase output, the detrended signals are zero-padded to N_d samples prior to correlation. For the GFRP AP dataset, $N_d \in \{210, 420, 630, 840, 1000, 2048, 4096\}$ is explored. For the GFRP flux dataset, $N_d \in \{160, 640, 1024, 2048\}$ is used.

IV. Results

A. Cross-Correlation

Fig. 2 shows the cross-correlation of pixel $(70, 190)$ with the reference in the CFRP LFM dataset. A well-defined main lobe is visible at the centre of the correlation, flanked by decaying side lobes characteristic of an LFM waveform.

Fig. 3 compares the cross-correlation signals for twelve defect pixels against a sound reference pixel in the GFRP flux dataset. While most defect pixels exhibit broadly similar main-lobe shapes, several show elevated pre-peak side-lobe energy and asymmetric lobe structure compared to the sound pixel, consistent with altered heat diffusion over subsurface anomalies.

Fig. 4 shows the FFT phase of the correlation window for two defect pixels and one sound pixel in the GFRP AP dataset. All three profiles are nearly indistinguishable: each

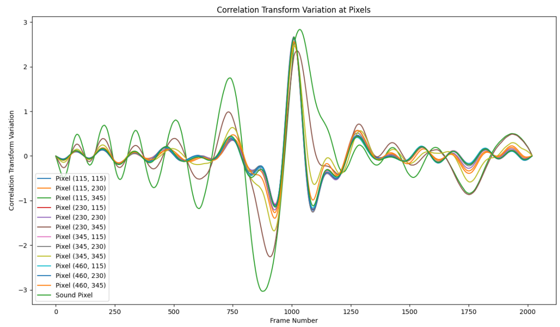

Fig. 3. Cross-correlation signals for twelve defect pixels and one sound pixel in the GFRP flux dataset. Some defect pixels display elevated pre-peak side lobes not present in the sound pixel.

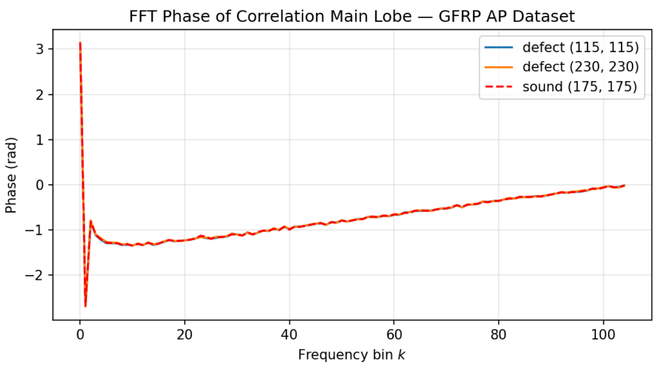

Fig. 4. FFT phase of the extracted correlation window for two defect pixels — $(115, 115)$ and $(230, 230)$ — and the sound pixel at $(175, 175)$ in the GFRP AP dataset. All three profiles overlap, indicating that active pulsed excitation does not produce phase-separated signatures with this pipeline.

exhibits a sharp peak near $+\pi$ at bin 1, decays sharply to approximately -2.5 rad at bin 2, and then rises smoothly toward zero across the remaining bins. The absence of phase separation between defect and sound pixels indicates that the active pulsed excitation does not imprint a frequency-localised phase signature in the extracted correlation window, in contrast to the results obtained with flux excitation below.

B. FFT Phase of the Correlation Main Lobe

Fig. 5 shows the FFT phase of the main lobe for three defect pixels and one sound pixel in the GFRP flux dataset. Two of the defect pixels — at $(115, 230)$ and $(115, 345)$ — produce nearly identical phase profiles: the phase rapidly saturates near $+\pi$, then drops discontinuously to $-\pi$ around frequency bin 80 before decaying. The third defect pixel at $(115, 115)$ exhibits a markedly slower, quasi-linear phase sweep across the full 160-bin range with a single wrap discontinuity. In contrast, the sound pixel at $(10, 10)$ shows high-frequency oscillations of large amplitude throughout, with no structured sweep. The three qualitatively distinct behaviours — fast-saturating, slow-sweeping, and rapidly oscillating — demonstrate that the FFT phase of the correlation main lobe separates material regions according to their thermophysical character.

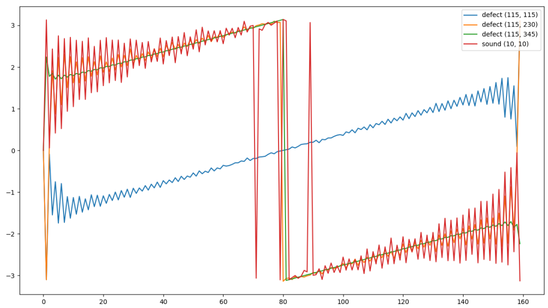

Fig. 5. FFT phase of the correlation main lobe for three defect pixels and one sound pixel in the GFRP flux dataset. Defect pixels at $(115, 230)$ and $(115, 345)$ (orange and green) share a fast-saturating profile; $(115, 115)$ (blue) exhibits a slow quasi-linear sweep; the sound pixel $(10, 10)$ (red) oscillates rapidly with no structured variation.

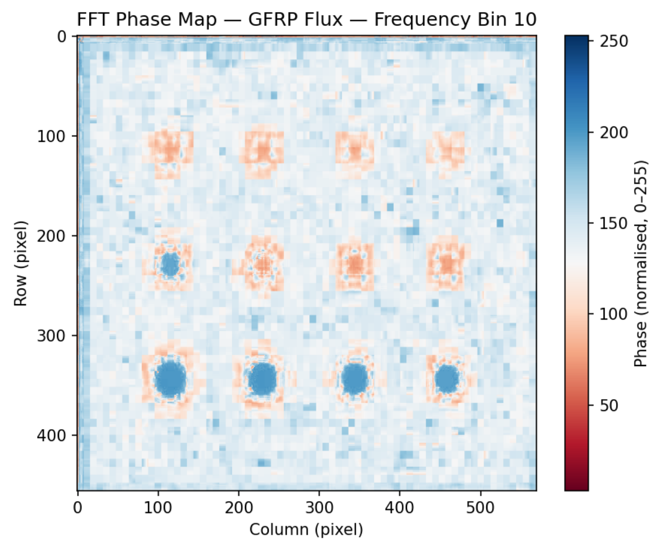

Fig. 6. Spatial FFT phase map at frequency bin 10 for the GFRP flux dataset. Circular anomalous regions (defects) are clearly separated from the surrounding sound material, demonstrating per-pixel spatial resolution of the pipeline.

C. Spatial Phase Map

Fig. 6 renders the per-pixel FFT phase at frequency bin 10 as a 2D spatial image for the GFRP flux dataset. Bin 10 was selected as the frequency index at which defect-to-sound phase contrast was visually maximised across the spatial map. Six to eight circular regions of elevated phase are clearly visible, arranged in a regular grid pattern consistent with the known subsurface defect layout of the specimen. The surrounding sound material produces a distinct, spatially uniform phase level. Phase values were mapped to the range $[0, 255]$ for video storage prior to display, so the colorbar reflects this normalised scale rather than absolute radians. This demonstrates that the pipeline generates spatially resolved defect maps directly from the per-pixel phase computation, without any segmentation or post-processing step.

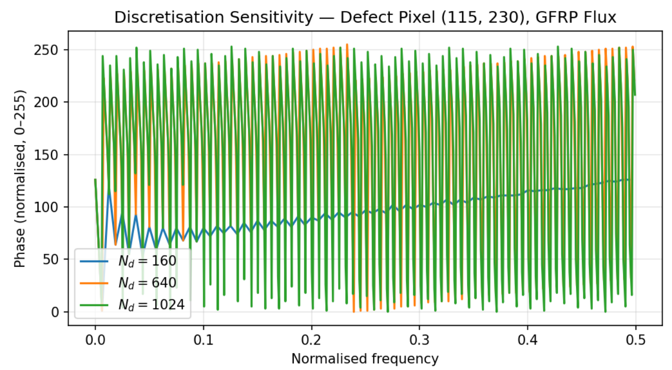

Fig. 7. FFT phase of defect pixel $(115, 230)$ in the GFRP flux dataset at three zero-padding levels. Increasing N_d refines frequency resolution but leaves the qualitative phase profile unchanged.

D. Discretisation Sensitivity

Fig. 7 shows the FFT phase of defect pixel $(115, 230)$ at zero-padding levels $N_d \in \{160, 640, 1024\}$ for the GFRP flux dataset. Phase values are read from the per-N_d output videos (normalised to $[0, 255]$ for storage), so the y-axis reflects this normalised scale. All three profiles exhibit the same oscillatory structure across the normalised frequency range; increasing N_d provides finer frequency resolution but does not alter the qualitative shape of the phase profile. This confirms that the discriminating features are stable across a wide range of padding choices, and that $N_d = 160$ (the natural main lobe length) already captures the essential phase behaviour.

V. Discussion

Linear detrending effectively removes bulk-heating drift, leaving the excitation-driven component of each pixel's thermal signal. Cross-correlation then measures the similarity of this component to the reference pixel's response as a function of temporal lag, enhancing defect contrast by suppressing spatially uniform background.

The FFT phase computed over the main lobe provides a frequency-domain signature that is qualitatively distinct between defective and sound regions in the GFRP flux dataset. The slow quasi-linear phase sweep observed for pixel $(115, 115)$ and the fast-saturating profiles of $(115, 230)$ and $(115, 345)$ suggest that defects at different locations — and potentially different depths — produce different phase responses. The rapidly oscillating phase of the sound pixel, by contrast, carries no structured trend. This structured vs. unstructured phase behaviour is the key discriminating observable, and the spatial phase map (Fig. 6) shows that it generalises across the specimen surface with sub-region spatial resolution.

The absence of phase separation in the GFRP AP dataset (Fig. 4) reflects the waveform structure of active pulsed excitation. A short broadband thermal pulse produces a diffuse thermal response in which defect-induced diffusivity differences manifest as small amplitude perturbations rather than frequency-localised phase shifts. The cross-correlation of two such pulse responses resembles an autocorrelation of a broad impulse response function, which is symmetric and concentrated at zero lag regardless of local diffusivity. As a result, the phase of the correlation window is dominated by the excitation waveform rather than material properties, preventing defect discrimination. Chirp-based (LFM) and flux-based excitations sweep frequency over time and therefore impart structured time-frequency content; this is what enables the per-frequency phase diversity exploited by the pipeline.

The discretisation experiments (Fig. 7) confirm that the qualitative phase profile is stable across a wide range of zero-padding levels, so practitioners can choose N_d for computational convenience without sacrificing discriminating power. A systematic quantitative analysis relating disc level N_d to defect depth is left for future work, building on depth-resolved pulse compression approaches developed for GFRP characterisation [4].

VI. Conclusion

We have presented and evaluated a signal processing pipeline for active thermographic NDT of FRP composites. The pipeline — linear detrending, full cross-correlation with a sound reference pixel, main lobe extraction, and FFT phase computation — is applied to CFRP (LFM excitation) and GFRP (active pulsed and flux excitation) datasets [3]. For the GFRP flux dataset the FFT phase of the correlation main lobe produces qualitatively distinct signatures that clearly separate defective from healthy material regions and, when rendered as a 2D spatial map, reveals the subsurface defect layout without additional segmentation. The pipeline's discriminating power is sensitive to excitation waveform: active pulsed excitation does not produce phase separation, whereas the GFRP flux dataset — which carries structured time-frequency content — does. The CFRP LFM cross-correlation displays expected main-lobe structure consistent with chirp-based excitation; per-pixel phase discrimination for that dataset is reserved for future work. Future work will establish quantitative relationships between the phase profile shape and defect depth, and will investigate adaptive reference-pixel selection to improve robustness.

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